

JARVIS Vision Assistant

Student

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Course

ITAI 1378 - Computer Vision and AI

Tier

Tier 1

Tier 1 was selected because this project uses a pretrained computer vision model to create a functional object-detection application without requiring a custom model to be trained from scratch.

Problem Statement

People cannot always quickly identify or keep track of objects in their surroundings. Computer vision can help automate this process by analyzing images and identifying common objects.

This project explores how a vision-based AI assistant can recognize objects and provide useful information about what it sees.

Solution Overview

JARVIS Vision Assistant will take an image as input and use a pretrained object-detection model to identify common objects.

The system will display detected objects with labels and bounding boxes.

Input to Object Detection Model to Object Labels to Results

Technical Approach

- **Computer Vision Technique:** Object Detection
- **Model:** Pretrained YOLO model
- **Framework/Tools:** Python, Ultralytics YOLO, OpenCV, Google Colab, GitHub

YOLO was selected because it can efficiently identify multiple objects within the same image.

Data Plan

The project will use approximately 30-50 test images. Images will include student-collected photographs and appropriate public/sample images containing everyday objects such as laptops, cell phones, bottles, chairs, backpacks, cars, and people.

The pretrained YOLO model will provide the object labels.

Success Metrics

Primary Metric

Target: At least 85% of clearly visible supported objects in the selected test set should be correctly identified.

Secondary Metric

Target: Process each image in under 2 seconds in the main testing environment.

Milestone Plan

Phase	Goal	Target
Blueprint	Complete project plan	Week 10
First Working Demo	Run YOLO on sample images	Week 11
Make It Yours	Add personal test images and application logic	Weeks 12-13
Improve and Measure	Test accuracy and processing speed	Week 14
Package and Present	Complete demo, README, slides, and presentation	Week 15

Risks and Plan B

Risk 1

The model may fail to correctly recognize certain objects.

Plan B: Use clearer images, better lighting, supported object categories, and appropriate confidence settings.

Risk 2

Real-time camera detection may be too complicated or slow.

Plan B: Use uploaded still images while maintaining the complete computer-vision pipeline.

Resources and Estimated Cost

- Google Colab
- Python
- Ultralytics YOLO
- OpenCV
- GitHub
- **Estimated Cost:** \$0 using free/open-source software and free compute resources where available.

AI Usage

AI assistance is documented in docs/AI_usage_log.md.